

Escaping the Curse of Dimensionality in the Kinetic Theory of Ordered Fluids via PETSc DMSwarm



Mathematical
Institute

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PDESofT 2026, TU Wien, 13th–15th July 2026



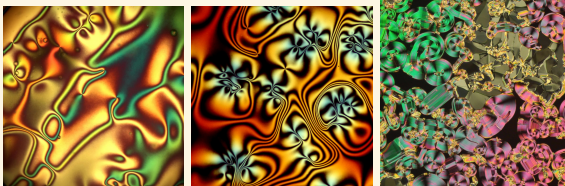
Oxford
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The constituents carry *orientation*: the natural description is a **kinetic equation** on a high-dimensional phase space.

KINETIC THEORY & THE CURSE OF DIMENSIONALITY

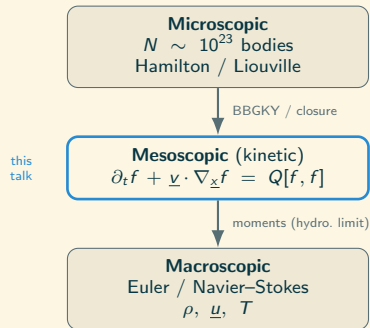
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KINETIC THEORY: THE MESOSCOPIC BRIDGE

A fluid is $N \sim 10^{23}$ constituents: following every trajectory (Liouville on a $6N$ -dimensional phase space) is hopeless.

Statistical mechanics replaces them by a **distribution function** $f(t, \underline{x}, \underline{v})$ – the expected particle density in phase space.

The **kinetic** (Boltzmann) equation evolves f at the *mesoscopic* level, keeping the statistics a hydrodynamic model averages away.



K. Huang, Statistical Mechanics; M. Kardar, Statistical Physics of Particles; C. Cercignani, The Boltzmann Equation.

THE MESOSCOPIC VIEWPOINT

We follow the **one-particle distribution function** $f(t, \underline{q}, \underline{p}) \geq 0$ over generalized coordinates $\underline{q}$ and conjugate momenta $\underline{p}$:

$$\text{simple gas: } (\underline{q}, \underline{p}) = (\underline{x}, \underline{v}), \quad \text{ordered fluid: } (\underline{q}, \underline{p}) = ((\underline{x}, \theta), (\underline{v}, \omega)).$$

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It transports f along the Hamiltonian characteristics ($\dot{\underline{q}} = \partial_{\underline{p}} H$, $\dot{\underline{p}} = \text{force}$):

$$\partial_t f + \underbrace{\dot{\underline{q}} \cdot \nabla_{\underline{q}} f}_{\text{free streaming}} + \underbrace{\dot{\underline{p}} \cdot \nabla_{\underline{p}} f}_{\text{forces}} = \underbrace{Q[f, f]}_{\text{collisions}}.$$

Boltzmann: $\dot{\underline{q}} = \underline{v}$, $\dot{\underline{p}} = \underline{F}$. CFMZ: $\dot{\underline{q}} = (\underline{v}, \omega)$, $\dot{\underline{p}} = (\underline{F}, \mathcal{V})$ – the torque $\mathcal{V}$ drives ω .

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 All macroscopic observables (density, momentum, temperature, **order parameters**) are moments of f in the microscopic variables.

C. Cercignani, The Boltzmann Equation and Its Applications; *C. Villani*, Handbook of Math. Fluid Dyn.

THE CURSE OF DIMENSIONALITY

A deterministic (grid) solver puts n points per axis on a d -dimensional phase space, coupled through a local **stencil**:

$$\# \text{ degrees of freedom} = n^d,$$

growing **exponentially** in d . For ordered fluids d is large *before* space is even resolved.

With $n = 64$: a 6-D grid already needs $\sim 7 \times 10^{10}$ degrees of freedom; a 10-D grid needs $\sim 10^{18}$.

Model	variables	d
Boltzmann gas (2D)	$\underline{x}, \underline{v}$	4
CFMZ needles (homog.)	$\underline{v}, \theta, \omega$	4
CFMZ needles (space)	$\underline{x}, \underline{v}, \theta, \omega$	6
3D nematic LC	$\underline{x}, \underline{v}, \underline{n} \in \mathbb{S}^2, \underline{\omega}$	~ 10

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Grid quadrature error $\sim (\# \text{DOF})^{-p/d}$: the more dimensions, the worse each unknown pays off.

PARTICLES ESCAPE THE CURSE

Represent f by an **empirical measure** of N samples (particles):

$$f(t, \underline{z}) \approx \frac{1}{N} \sum_{p=1}^N \delta(\underline{z} - \underline{z}_p(t)) .$$

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- ✓ cost per step $\mathcal{O}(N)$ (cell-local);
 - ✓ memory $\mathcal{O}(N)$, a few fields/particle;
 - ✓ trivially parallel over particles.
- ▶ statistical noise $\sim 1/\sqrt{N}$;
 - ▶ **contain & advance** the particles;
 - ▶ **handle the collisions** (Nanbu, BGK).

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 G. Dimarco & L. Pareschi, Numerical methods for kinetic equations, *Acta Numerica* 2014;
 L. Pareschi & G. Russo, Monte Carlo methods for the Boltzmann equation, *Trans. Theory Stat. Phys.*
 2000.

TWO KINETIC MODELS, ONE ENGINE

Rarefied gas – Boltzmann

$$\partial_t f + \underline{v} \cdot \nabla_{\underline{x}} f = Q[f, f]$$

Maxwell molecules; $f(t, \underline{x}, \underline{v})$.

Sod shock tube, flow past a cylinder.

Ordered needles – CFMZ

$$\begin{aligned} \partial_t f + \underline{v} \cdot \nabla_{\underline{x}} f + \omega \partial_{\theta} f \\ + G[f] \cdot \nabla_{\underline{v}} f + F[f] \partial_{\omega} f = Q[f, f] \end{aligned}$$

rigid rods; $f(t, \underline{x}, \underline{v}, \theta, \omega)$; Onsager force $G[f]$, torque $F[f]$; homogeneous $\Rightarrow$ drop $\underline{v} \cdot \nabla_{\underline{x}}$, $G[f]$.
Isotropic–nematic transition.

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Both share the **transport** + **collision** structure, so a single DSMC engine built on DMSwarm solves both.



CFMZ kinetic model: *J. A. Carrillo, P. E. Farrell, A. Medaglia, ~, arXiv:2508.10744 (2025).*

PETSC DMSWARM

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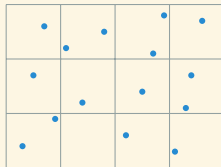
WHAT IS DMSWARM?

A DMSwarm is a **parallel bag of particles** attached to a background *cell* DM:

- ▶ each particle carries user-**registered fields** (velocity, orientation, ...);
- ▶ particles are **located in cells** of the background mesh;
- ▶ cheap insertion/deletion; **MPI migration** between ranks;
- ▶ interoperates with the whole PETSc/ DMDA ecosystem.

Exactly the data structure DSMC needs: a distributed particle store with fast cell look-up for local collisions.

particles = DMSwarm



background DMDA (cells)



DMSWARM FOR DSMC

```

swarm = PETSc.DMSwarm().create(comm=comm)
swarm.setType(PETSc.DMSwarm.Type.PIC)      # particle-in-cell
swarm.setCellDM(dm)                        # background DMDA

swarm.registerField("velocity",             2, dtype=PETSc.RealType)
swarm.registerField("orientation",         1, dtype=PETSc.RealType) # theta
swarm.registerField("angular_velocity",    1, dtype=PETSc.RealType) # omega
swarm.finalizeFieldRegister()
swarm.setLocalSizes(nlocal, N)              # local size, migration buffer

```

Boltzmann: 2-D spatial DMDA; particles stream through physical space and migrate across ranks.

CFMZ: 1-D periodic DMDA over $\theta \in [0, 2\pi)$; space-homogeneous, so reductions are global Allreduces.

LOCALITY & PARALLELISM

After free streaming, particles that crossed a rank boundary are **migrated**:

```
# advect, then hand off to owning rank
X[:] += V * dt
swarm.migrate(remove_sent_points=True)
```

Collisions are **cell-local**: bucket particles by their cell id.

Collisions only couple nearby particles, so each rank **buckets its particles by cell** (a sort on the swarm's cell index) and draws collision pairs *within* each cell:

```
cells = get_particle_cells(swarm)
for c, idx in group_by_cell(cells):
    collide_pairs(idx)    # local to cell c
```

Cost stays $\mathcal{O}(N)$ and each collision respects the local velocity distribution – no global interactions.

THE DSMC ALGORITHM

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OPERATOR SPLITTING

Each step splits the transport (Hamiltonian) part from the collision (stochastic) part by second-order **Strang splitting**:

$$f^{n+1} = \mathcal{T}_{\Delta t/2} \circ \mathcal{Q}_{\Delta t} \circ \mathcal{T}_{\Delta t/2} f^n.$$

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For CFMZ the transport itself is a Störmer–Verlet **drift–kick–drift**:

$$\mathcal{T}_{\Delta t} = \mathcal{D}_{\Delta t/2} \circ \mathcal{K}_{\Delta t} \circ \mathcal{D}_{\Delta t/2}, \quad \begin{cases} \mathcal{D}: & \theta += \omega \Delta t \\ \mathcal{K}: & \omega += F[f](\theta) \Delta t \end{cases}$$

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G. Strang, On the construction and comparison of difference schemes, SIAM J. Numer. Anal. 1968.

THE COLLISION SUBSTEP

Nanbu

Maxwell molecules: draw $M_{\text{col}} = \lfloor \nu N \Delta t / 2 \rfloor$ pairs uniformly; conserve momentum and energy; randomise scattering.

Bird NTC

Hard needles: accept a pair with prob. w/ν_{max} , $w = |\underline{g} \cdot \underline{n}| L |\sin \Delta \theta|$ – near-parallel rods almost never collide.

BGK

Resample each cell from its local Maxwellian; rescale to conserve momentum & energy exactly.

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All three act *inside a cell* on the bucketed particle indices – the same DMSwarm cell-list drives every kernel.



K. Nanbu, J. Phys. Soc. Japan 1980;

G. A. Bird, Molecular Gas Dynamics and the Direct Simulation of Gas Flows, 1994;

Bhatnagar–Gross–Krook, Phys. Rev. 1954.

THE NANBU-BABOVSKY ALGORITHM

A first-order discretisation of the Maxwell-kernel collision is the *convex combination*

$$f^{n+1} = \left(1 - \frac{\Delta t}{\tau}\right) f^n + \frac{\Delta t}{\tau} C^+[f^n, f^n],$$

a probability density read particle-wise. Per step:

1. $N_c = \text{Sround}(N\Delta t/2\tau)$ interacting pairs;
2. pick N_c pairs (i, j) uniformly at random;
3. draw impact normal $\underline{n}(\psi)$, contact arm ℓ ; form the impulse J ; update $(\underline{v}_i, \omega_i), (\underline{v}_j, \omega_j)$;
4. leave the rest unchanged; θ is never changed by a collision.

Ingredients

- ▶ Sround: stochastic rounding, $\mathbb{E}[\text{Sround}(x)] = x$;
- ▶ elastic ($e_v = e_\omega = 1$) $\Rightarrow$ conserves momentum, angular momentum, energy;
- ▶ **hard-needle** kernel: keep the convex form via Bird's acceptance $w/w_{\max}$ (null collisions otherwise).

All pair selection and updates are **cell-local** on the DMSwarm.

RAREFIED GAS DYNAMICS

4

SOD SHOCK TUBE

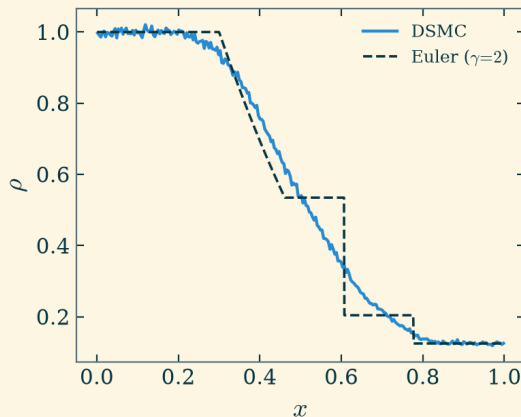
Left / right initial states

$$\rho_L = 1, \quad T_L = 1.0$$

$$\rho_R = 0.125, \quad T_R = 0.8$$

- ▶ domain $x \in [0, 1]$, 256 cells;
- ▶ $\sim 10^7$ particles, density ratio realised by particle count;
- ▶ free streaming + elastic wall reflection + per-cell collisions.

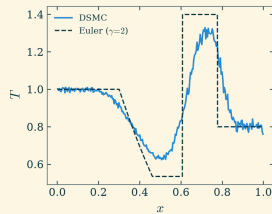
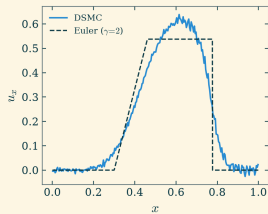
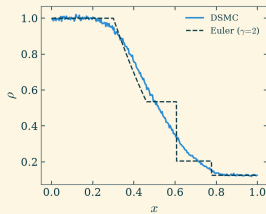
We compare **Nanbu** vs **BGK** collisions at $t = 0.2$.



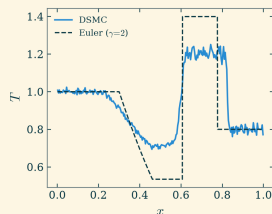
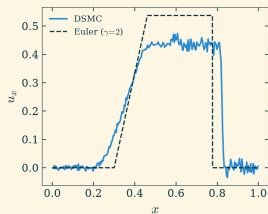
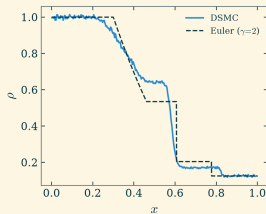
density $\rho(x)$ at $t = 0.2$ (Nanbu)

SOD SHOCK TUBE: NANBU VS BGK AT THE SAME Kn

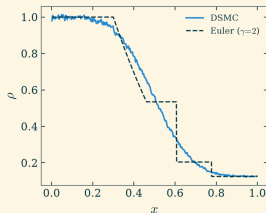
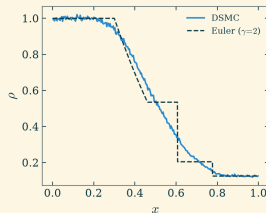
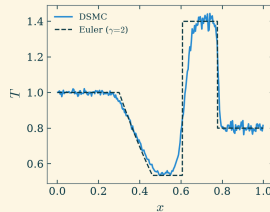
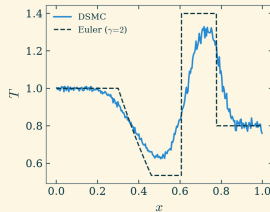
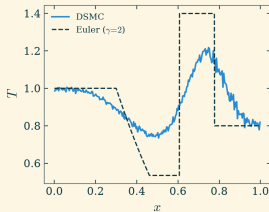
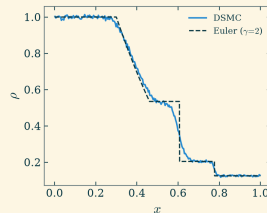
Nanbu



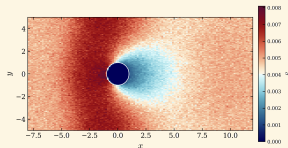
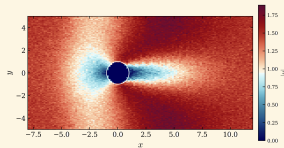
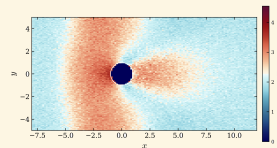
BGK

 ρ u_x T

SOD SHOCK TUBE: VARYING THE KNUDSEN NUMBER (NANBU)

 $Kn \approx 0.1 \quad (\nu = 10)$

 $Kn \approx 0.01 \quad (\nu = 100)$

 $Kn \approx 0.001 \quad (\nu = 1000)$


FLOW PAST A CYLINDER – $Re \approx 15$

density ρ speed $|\underline{u}|$ temperature T

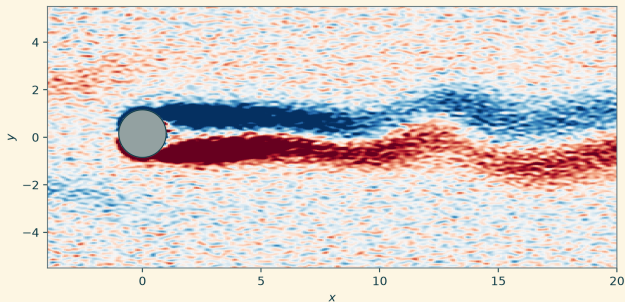
Setup: uniform Maxwellian + drift $u_\infty = 1.5$ on $[-8, 12] \times [-5, 5]$, 128^2 cells, $\sim 10^7$ particles.

- ▶ periodic in x ; specular reflection on walls & cylinder;
- ▶ $Kn = \lambda/R \approx 0.1$, viscosity $\mu \sim T/(m\nu)$,
 $Re = u_\infty R/\mu \approx 15$.

A steady laminar wake (shedding only sets in near $Re \sim 40$):

- ▶ stagnation heating upstream, cooling in the wake;
- ▶ a density shadow downstream;
- ▶ the flow accelerates around the sides, symmetric top–bottom.

FLOW PAST A CYLINDER AT $Re \approx 270$



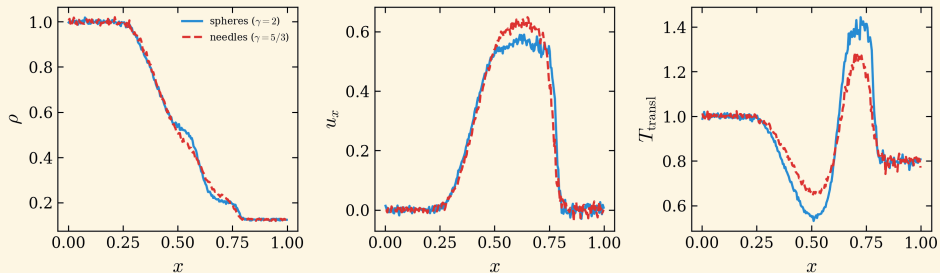
vorticity $\omega_z(x, y)$ at $t = 30$ (open inflow/outflow BC, smoothed)

Above a critical Reynolds number the wake behind a bluff body goes unstable and sheds a periodic train of **alternating vortices** – the **von Kármán street** ($St \approx 0.2$). Here the two counter-rotating shear layers separate and **meander** downstream: the mode is switching on.

Parameters:

- ▶ 3.6×10^7 particles, 384^2 cells;
- ▶ $Re = u_\infty D / \mu \approx 270$, $u_\infty = 1.5$;
- ▶ open BC: reservoir inflow + absorbing outflow.

DOES THE CONSTITUENT'S SHAPE MATTER? SPHERES VS NEEDLES



Same Sod tube – *identical* ρ , p and translational T – run with **spheres** (2 translational DOF, $\gamma = 2$) and **needles** (2 translational + 1 rotational DOF, $\gamma = 5/3$) via the spatially-inhomogeneous CFMZ solver.

The needle's **translational** temperature rises *less* behind the shock ($T \approx 1.27$ vs 1.43): the rotational mode absorbs part of the compression heating (equation of state set by the internal DOF, if collisionally coupled, $I = mL^2/12$).

Relevant to **cold plasmas**: their rod-like constituents carry rotational degrees of freedom, so a point-sphere model mis-predicts the shock and energy balance.

KINETIC THEORY OF ORDERED FLUIDS

5

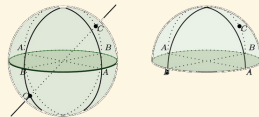
THE KINETIC MODEL OF ORDERED FLUIDS

Each constituent carries a position $\mathbf{x}$, momentum $\underline{p}$, an **order parameter** $\nu \in \mathcal{M}$ (a manifold with an $SO(d)$ action) and its conjugate momentum $\underline{\varsigma}$. A BBGKY hierarchy gives, for $f(\mathbf{x}, \nu, \underline{p}, \underline{\varsigma}, t)$,

$$\partial_t f + m^{-1} \underline{p} \cdot \nabla_{\mathbf{x}} f + \underline{\underline{B}}(\nu)^{-1} \underline{\varsigma} \cdot \nabla_{\nu} f + \mathcal{V} \cdot \nabla_{\underline{\varsigma}} f = \mathcal{C}[f, f],$$

with inertia tensor $\underline{\underline{B}}(\nu)$, mean-field force $\mathcal{V}$, collisions $\mathcal{C}$.

- ▶ $\mathcal{C}$ conserves mass, momentum, generalised angular momentum, energy (Noether);
- ▶ an **H -theorem** $\Rightarrow$ relaxation to a Maxwellian.



e.g. the nematic director manifold $\mathbb{RP}^2$ (3-D); the calamitic rod uses $\mathcal{M} = \mathbb{S}^1$



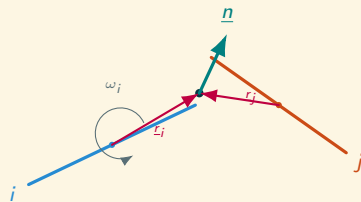
Kinetic theory: *J. A. Carrillo, P. E. Farrell, A. Medaglia*, ~ (companion work, 2025).

NEEDLES: THE CFMZ MODEL

The scheme is built for a *generic* order-parameter manifold $\mathcal{M}$; for calamitic (rod-like) molecules $\mathcal{M} = \mathbb{S}^1$ and each particle carries $(\underline{v} \in \mathbb{R}^2, \theta, \omega)$. A binary collision draws an impact normal $\underline{n}$ and a contact arm $\ell \in [0, L]$; the impulse

$$J = -\frac{\underline{V} \cdot \underline{n}}{\frac{2}{m} + (c_i^2 + c_j^2)/I}, \quad \underline{V} = (\underline{v}_i - \underline{v}_j) + \omega_i \underline{r}_i^\perp - \omega_j \underline{r}_j^\perp$$

updates both linear and angular velocities, conserving momentum, angular momentum and (elastic) energy.



two colliding rods: contact normal $\underline{n}$, contact arms $\underline{r}_i, \underline{r}_j$

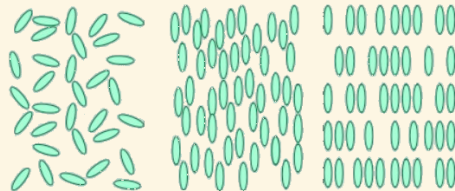
MEAN-FIELD ALIGNMENT & THE ONSAGER POTENTIAL

A pair potential $\mathcal{W}(\theta, \theta')$ generates the conservative torque

$$F[f](\theta) = -\partial_{\theta} (\mathcal{W} * \rho_{\theta})(\theta).$$

Three choices of increasing fidelity, all favouring alignment:

- ▶ quadratic: $F = -\alpha(\theta - \hat{\theta})$;
- ▶ Kuramoto: $F = -\sin(\theta - \hat{\theta})$;
- ▶ **Onsager** excluded volume $\mathcal{W} = L^2 |\sin(\theta - \theta')|$:
parallel rods have zero excluded volume, so it
drives *nematic* ordering.



liquid (no order) · **nematic** (orientation) ·
smectic (orientation + layers)

RECONSTRUCTING THE MEAN-FIELD FORCE

The force depends on the unknown f , so it is **reconstructed from the swarm at every step**:

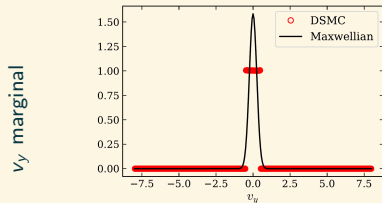
- ▶ mean-orientation forces (Kuramoto): the circular mean $\hat{\theta}$, cost $\mathcal{O}(N)$;
- ▶ general potential (Onsager): the density ρ_θ by **cloud-in-cell** deposition onto the DMSwarm cell grid, cost $\mathcal{O}(N_\theta^2)$, *independent of N* .

CIC is the classical particle-in-cell deposition (Hockney–Eastwood): each particle splits its weight between its two nearest cells.

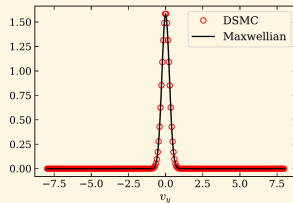
```
# CIC deposition on the theta cell-grid
rho = cic_density(theta)           # Allreduce
Phi = dtheta * (W_mat @ rho)       # W * rho
k   = (theta // dtheta).astype(int) % bins
F   = L**2 * (Phi[k]
              - Phi[(k+1) % bins]) / dtheta
```

The *continuous* dynamics conserves $\mathcal{E} = E_{\text{kin}} + E[\rho]$, which the symplectic step then tracks with a small, bounded drift (next slides).

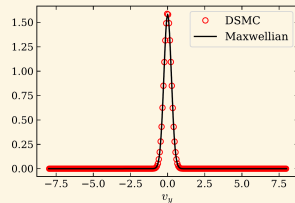
VALIDATION: THERMALISATION & THE H -THEOREM



$t = 0$: bimodal



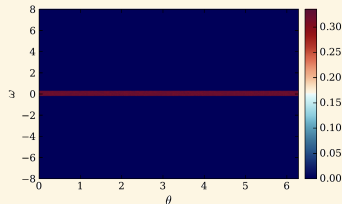
$t = 5$



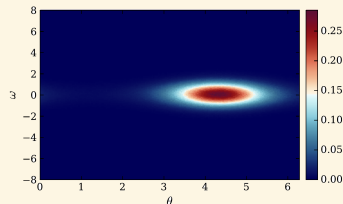
$t = 50$: Maxwellian

A non-equilibrium (bimodal) datum relaxes onto the Maxwellian predicted by the H -theorem; DSMC markers (○) match the exact equilibrium (line).

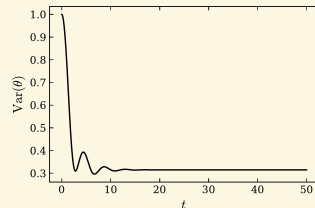
MEAN-FIELD ALIGNMENT: POLAR ORDER



(θ, ω) , $t = 0$: uniform



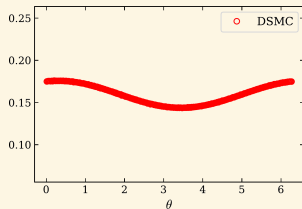
$t = 10$: one aligned lobe



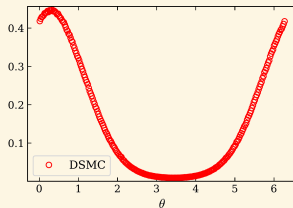
$\sigma^2(t)$: damped oscillations

The self-consistent torque concentrates (θ, ω) about the mean director; the **damped oscillations** of the order parameter are the signature of the mean-field coupling (contrast the two-lobed *nematic* Onsager state).

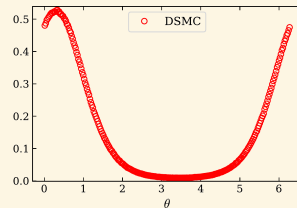
NUCLEATION OF ORDER



$\theta, t = 0$: perturbed isotropic



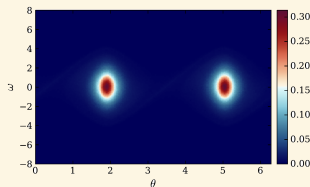
$t = 5$: cluster nucleates



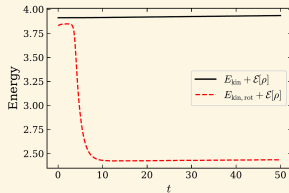
$t = 10$: ordered

Starting from a slightly perturbed isotropic datum, the Kuramoto dynamics drains the residual isotropic band into the nucleating cluster – symmetry breaking, captured particle-by-particle.

ENERGY CONVERSION & SYMPLECTIC TRANSPORT



(θ, ω) : nematic lobes form

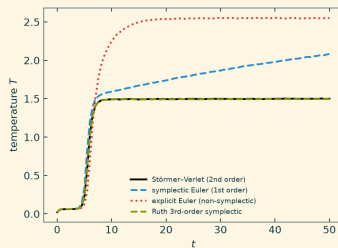
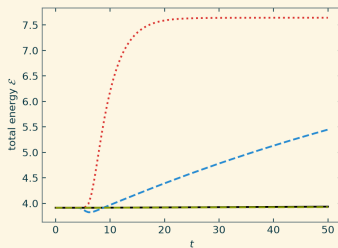
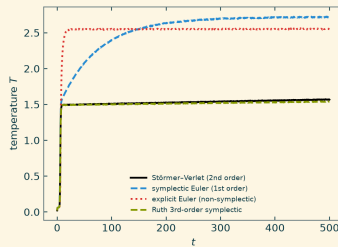
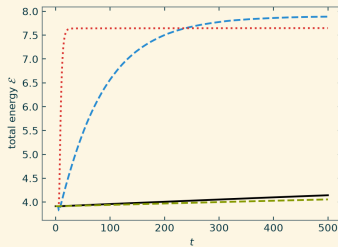


total energy conserved

As the rods align, interaction energy is converted into kinetic energy and the temperature rises.

The total $\mathcal{E} = E_{\text{kin}} + E[\rho]$ (solid) is **conserved**; the second-order **symmetric** Störmer–Verlet transport keeps the drift $< 1\%$ – a lower-order step drifts and drowns the physical effect (next slide).

WHY SECOND ORDER? A TIME-STEPPER EXPERIMENT

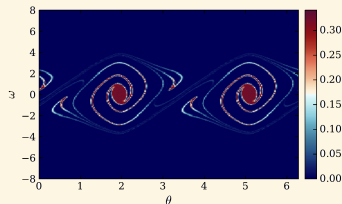


DSMC ($I=1$, $L=\sqrt{12}$), Strang-split transport $\circ$ collision – energy drift ($t=500$):

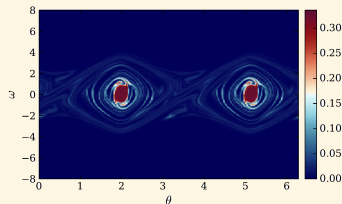
- ▶ **explicit Euler** +95%, **symplectic Euler** +102%: both 1st-order transports **blow up** ($T \rightarrow 2.6$);
- ▶ **Störmer-Verlet** (2nd): holds the plateau, +6%;
- ▶ **Ruth 3rd-order symplectic**: the best, +4%.

The collision breaks the symplectic structure each step, so a **first-order transport** error compounds and spuriously heats the system to $T \approx 2.6$. Only a **2nd-order or higher symmetric transport** tracks the conserved energy.

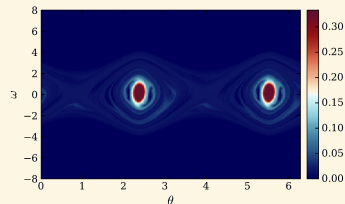
DIOCOTRON STYLE INSTABILITIES



$t = 5$



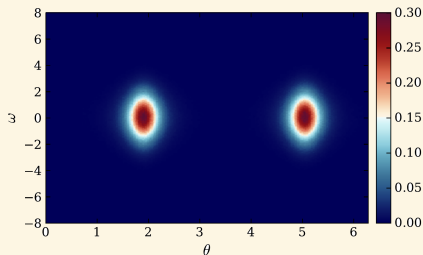
$t = 15$



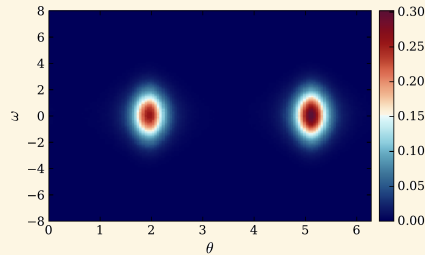
$t = 35$

Switch the collisions off and the (θ, ω) density rolls up into ever-finer filaments. Particles transport them **without artificial diffusion**; a phase-space grid would smear them the moment they drop below the mesh size.

ROBUSTNESS TO THE COLLISION KERNEL



Maxwell kernel: all pairs equally likely



hard-needle: $w = |\underline{g} \cdot \underline{n}| L |\sin \Delta\theta|$, Bird's method

The hard-needle cross-section vanishes for parallel rods – a candidate pair collides with probability $w/w_{\max}$ (null collisions otherwise) – yet the ordering is **unchanged**. Same DMSwarm engine, only the acceptance rule differs.

THE ISOTROPIC–NEMATIC TRANSITION

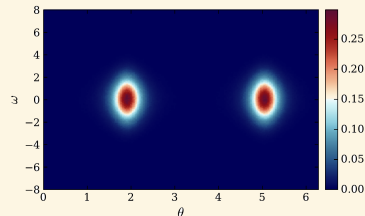
Head–tail symmetry ($\theta \sim \theta + \pi$): orientations live on the real projective line $\mathbb{RP}^1$, identified with $\mathbb{S}^1$ by angle doubling $\theta \mapsto 2\theta$. The order parameter is the circular variance

$$\sigma^2 = 1 - R_2, \quad R_2 = |\langle e^{2i\theta} \rangle|, \quad \begin{cases} \sigma^2 \rightarrow 1 & \text{isotropic} \\ \sigma^2 \rightarrow 0 & \text{nematic} \end{cases}$$

(R_2 is the 2-D nematic scalar order parameter.) Linear stability of the isotropic state $\rho_0 = \frac{1}{2\pi}$ under the Onsager self-consistency $\rho \propto \exp(-(\mathcal{W} * \rho)/T)$ gives the **spinodal**

$$\alpha_c = L^2/T_c = \frac{3\pi}{2}, \quad T_c = \frac{2L^2}{3\pi} = \frac{8}{\pi} \quad (L = \sqrt{12}),$$

with unstable mode $\cos(2(\theta - \theta_0))$: a **second-order** transition.



equilibrium (θ, ω) : two head–tail symmetric lobes



L. Onsager (1949); Carrillo, Farrell, Medaglia & ~, arXiv:2508.10744.

ISOTHERMAL ENSEMBLES: THE THERMOSTAT

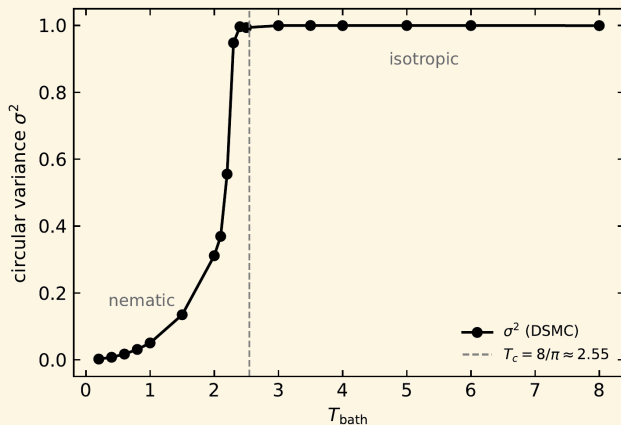
So far the run is **microcanonical**: $\mathcal{E}$ is fixed and the temperature rises as order forms. To scan the phase diagram at fixed T we add an **Andersen thermostat**: with probability $\nu_{\text{bath}} \Delta t$ each particle resamples

$$\underline{v} \sim \mathcal{N}(0, \sqrt{T_{\text{bath}}/m}), \quad \omega \sim \mathcal{N}(0, \sqrt{T_{\text{bath}}/I_3}),$$

with θ untouched, once per step after the collisions.

This is the BGK bath term $Q_{\text{bath}} = \nu_{\text{bath}}(\mathcal{M}_{T_{\text{bath}}} - f) \Rightarrow$ the **exact NVT ensemble** $\rho \propto e^{-(\mathcal{W}^*\rho)/T_{\text{bath}}}$.

THE PHASE DIAGRAM

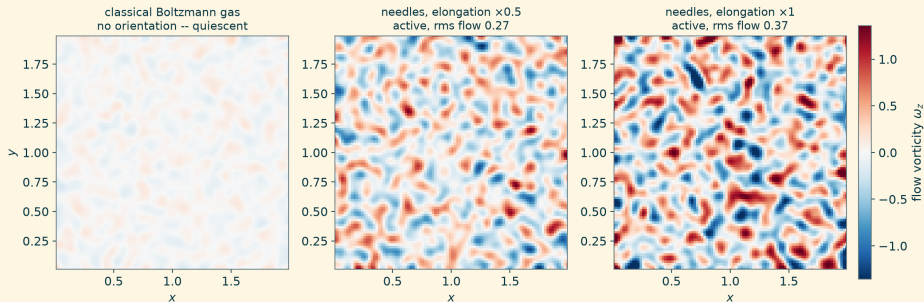


A BGK/Andersen thermostat fixes T_{bath} ; scanning it and recording σ^2 at equilibrium traces the phase diagram:

- ▶ σ^2 vanishes *continuously* at $T_c = 8/\pi \approx 2.55$;
- ▶ nematic below, isotropic above – a **second-order** transition;
- ▶ the mean-field spinodal $\alpha_c = 3\pi/2$ recovered from particles.

$N = 4 \times 10^5$ per run, $\nu_{\text{bath}} = 4$; agrees with Landau–de Gennes in 2-D.

ACTIVE NEMATIC TURBULENCE



Flow vorticity, one DMSwarm engine. **Gas**: structureless – quiescent. **Needles** self-propel along their axis, so activity converts orientation into **turbulent flow**; *more elongated* rods (Onsager coupling $\propto L^2$) drive it **harder** (rms $0.27 \rightarrow 0.37$).

A structureless gas cannot be stirred this way instead orientational **activity** is encompassed in the **Leslie–Ericksen** model, and rod elongation amplifies it.

CONCLUSIONS

6

CONCLUSIONS

- ✓ Ordered fluids **can be treated from a kinetic point of view**: their phase space is high-dimensional, and deterministic grids hit the **curse of dimensionality**.
- ✓ **Particle DSMC** escapes it – Monte-Carlo error $\sim 1/\sqrt{N}$, *independent of d* – and transports fine phase-space structure without artificial diffusion.
- ✓ **PETSc DMSwarm** supplies the parallel particle container (fields, MPI migration, cell-lists); a **symplectic** transport preserves the total energy, exposing the interaction-to-kinetic energy conversion that drives ordering.
- ✓ One engine delivers **Sod shocks**, an unsteady **cylinder wake**, and the second-order **isotropic–nematic transition** at $T_c = 8/\pi$.



J. A. Carrillo, P. E. Farrell, A. Medaglia, ~, arXiv:2508.10744 (2025). Code:
github.com/UZerbinati/dsmc

THANK YOU!

Escaping the Curse of Dimensionality in the
Kinetic Theory of Ordered Fluids via PETSc DMSwarm

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